

LXCE on ThinkSystem V3 servers

New features and enhancements

The Lenovo logo is positioned in the top right corner of the slide. It consists of the word "Lenovo" in a white, sans-serif font, oriented vertically. The text is set against a rectangular background that features a vertical color gradient, transitioning from a light green at the top to a light blue at the bottom.

Lenovo

Tool overview

LXCE has the following new features and enhancements for the ThinkSystem V3 platform:

- Version update to support ThinkSystem V3
 - OneCLI: 4.0.1
 - UpdateXpress: 4.0.1
 - BoMC: 13.0.1
- New OS support – Windows 10 and 11
 - OneCLI Binary
 - Invgy_utl_lxce_oneclixxx-4.0.1_winsrv_x86-64.zip
 - Invgy_utl_lxceb_oneclixxx-4.0.1_winsrv_x86-64.exe
 - UpdateXpress Binary
 - Invgy_utl_lxce_uxxxx-4.0.1_windows_x86-64.zip

Always check the Lenovo XClarity Essentials OneCLI (OneCLI) website for the latest OneCLI information and User Guide.

- <https://datacentersupport.lenovo.com/us/en/documents/Invo-tcli>

Major OneCLI software feature updates

OneCLI has the following feature updates for the ThinkSystem V3 platform:

- OneCLI bundle update
- OneCLI remote update
- OneCLI out-of-band (OOB) RAID on VROC and Microchip RAID controllers
- OneCLI one time boot configuration
- Miscellaneous new OneCLI functions

OneCLI – Bundle update

The new bundle update option comes with a new `--bundle` parameter and has the following features:

- Update firmware with a bundled package or packages
- OneCLI will send a .zip bundle pack with one or more firmware components to XCC to initiate the firmware update process

A new optional `-applytime` parameter has the following three values:

- `OnReset` - default for in-band (IB) mode
- `Immediate` - default for OOB mode
- `OnStartUpdateRequest` - users start the update process manually

The new manual staged update option works with the `checktask` and `startstaged` commands .

- `Checktask` - check staged bundle task status
- `Startstaged` - start staged update task

Bundle update in UpdateXpress

There is a new **Update firmware with bundles package(s)** option in the **Update Execution** section.

Lenovo XClarity Essentials UpdateXpress v4.0.0 - 01o

1. Welcome

2. Task

3. Setting

4. Update Location

5. Update Type

6. Update Comparison

7. Update Execution

8. Finish

Running updates

☒ Update firmware with bundled package(s). This checkbox and its suboptions only support XCC2.

└ Apply time

Immediate ▾

For "Immediate" Specifies it while you want conduct update immediately.

Begin Update

Lenovo XClarity Update Manager bundle package

The Lenovo XClarity Update Manager (LXUM) bundle package is a new core firmware component which allows XCC to boot a Bare Metal Update (BMU) environment to carry out iBMU updates for non-agentless I/O firmware like HDD and Persistent Memory (PMems).

For example: *lnvgy_fw_lxum_eal501c-1.00_anyos_comp*

Manual staged update example

In this example, the `--bundle` and `--applytime` parameters are used with the `OnStartUpdateRequest` to perform **value**.

```
OneCli.exe update flash --bmc "USERID:PASSWORD@10.241.136.145" --  
bundle --dir D:\onecli\egs\7d75-shanghai-bundle --applytime  
OnStartUpdateRequest --includeid lnvgy_fw_drvwn_eal301o-  
4.00_anyos_comp
```

In the screen capture shown, you can see Update task staged to BMC successfully at the prompt, and the task ID is also given:

```
Update task staged to BMC successfully, task id is: 82c05d24-da56-4e60-a112-7be807dca383
```

Flash Results:

No.	Updatable Unit	Slot	Need Reboot	Result	Update ID
1	LXPM Windows Drivers	N/A	Flashing requires additional command	Staged	lnvgy_fw_drvwn_eal301o-4.0 0_anyos_comp

Then use the `startstaged` command to execute the update:

```
OneCli.exe update startstaged --bmc  
"USERID:PASSWORD@10.241.136.145"
```


Manual staged update – checktask

You can also use `checktask` command to check staged bundle task status before executing the `startstaged` command:

```
OneCli.exe update checktask -b "USERID:PASSWORD@10.241.136.145" --  
taskid 82c05d24-da56-4e60-a112-7be807dca383
```

```
Task state is: completed
```

```
Task associated update job is: JobR000043-Update
```

```
Query job JobR000043-Update state...
```

```
Query update unit information...
```

Job State:

ID	Percent	StartTime	EndTime	State
JobR000043-Update	100	2022-09-20 23:36:47	2022-09-20 23:37:50	Completed

Job Steps of JobR000043-Update:

No.	Update Unit	Percent	Slot	StartTime	EndTime	State	Update ID
0	LXPM Windows Drivers	100	N/A	2022-09-20 23:36:47	2022-09-20 23:37:46	Completed	lnvgy_fw_drvwn_eal30lo-4.00_anyos_comp.uxz

```
Save to D:\OneCli\bins\lnvgy_utl_lxce_onecli02c-4.0.0_winsrv_x86-64\update\OneCli-Update-Job-status.xml
```

```
Succeed.
```


Remote updates

OneCLI has a new feature to support remote updates from Windows to Windows platforms and from Windows to Linux platforms:

- From Windows to Windows

- `OneCli.exe update flash --remoteos lxce_user:LXCE2021_OneCli@WIN-08VH5AM4P9U --dir D:\OneCli\bins\fw`
- As shown in the screen capture, some steps should be completed first

- From Windows to Linux

- `OneCli.exe update flash --remoteos root:SYS2022health@10.240.199.19 --dir D:\dd_fw`
- Linux binary in the specified directory
- Root privileges are required

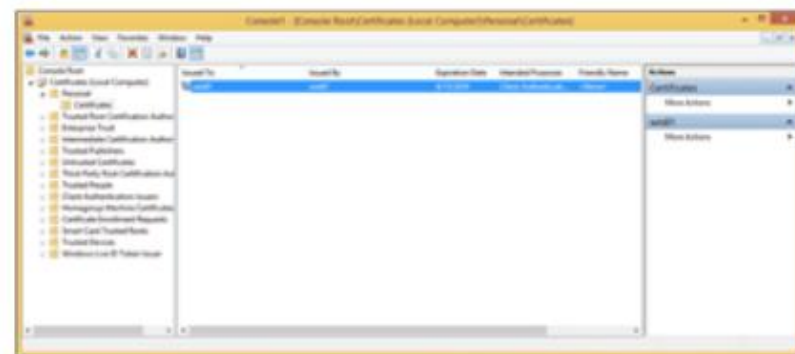
PowerShell 5 Remoting over HTTPS with a self-signed SSL certificate

Enabling HTTPS for PowerShell Remoting

On the remote computer

1. Create a self-signed ssl certificate with hostname or FQDN

```
$Cert = New-SelfSignedCertificate -CertstoreLocation Cert:\LocalMachine\My -DnsName  
<HostName | DomainName>
```



Certificate in MMC on the remote computer

2. Export the certificate to a file

```
Export-Certificate -Cert $Cert -FilePath C:\temp\cert
```

3. Enable PowerShell Remoting

```
Enable-PSRemoting -SkipNetworkProfileCheck -Force
```

Enable-PSRemoting also starts a WS-Management listener, but only for HTTP. If you want to, you can verify this by reading the contents of the WSMan drive:

```
dir wsman:\localhost\listener
```

OOB RAID on VROC and Microchip in OneCLI

- OOB RAID support on VROC – only supported on V3 servers with Intel processors and NVMe SSDs:

```
-OneCli.exe raid add --file  
  RAID_SW_new.ini --ctrl 3 -b  
  USERID:PASSWORD@10.240.195.209  
-File: RAID_SW_new.ini
```



```
[ctrl3-vol0]  
  disks=10,11  
  raid_level=1  
  vol_name=volume1  
  volume_size=600GB  
  strip_size=128K
```

- OOB RAID support on Microchip – this feature was first available with OneCLI 3.5 and is now also supported on the ThinkSystem V3 platform:

```
-OneCli.exe raid add --file  
  RAID_HW_new.ini --ctrl 2 -b  
  USERID:PASSWORD@10.240.195.209  
-File: RAID_HW_new.ini
```



```
[ctrl2-vol1]  
  disks=0,1  
  raid_level=1  
  vol_name=volume1  
  read_policy=0  
  write_policy=3  
  volume_size=300GB  
  strip_size=128K
```

OOB RAID on VROC and Microchip in UpdateXpress

In the RAID Configuration section of UpdateXpress, you can see that Microchip RAID adapters (5350-8i is shown as an example in the screen capture) and Intel VROC are supported.

Lenovo XClarity Essentials UpdateXpress v4.0.0 - 01o Active Machine Type: 7D75

1. Welcome
2. Task
3. Setting
4. RAID Configuration
5. Finish

RAID Configuration

Current RAID configuration information is displayed in the table below. Config it with function button.

Create Array Clear Controller Make Good Make JBOD Refresh

	Name	State	Capacity	Details	Serial Number
☐	Controller ThinkSystem RAID 940-32i				
☐	Non-RAID Drives				
☐	Drive 7	JBOD	232.886GB	SATA HDD	9XE0CDZS
☐	Drive 8	JBOD	279.397GB	SAS HDD	SAE00E13
☒	Controller ThinkSystem RAID 5350-8i				
☐	Non-RAID Drives				
☐	Drive 0	Unconfigured GOOD	558.912GB	SAS HDD	6WN0J3HG
☐	Drive 1	Unconfigured GOOD	558.912GB	SAS HDD	S0M7EPG4
☐	Drive 2	Unconfigured GOOD	136.732GB	SAS HDD	6TB1PNS6
☐	Drive 3	Unconfigured GOOD	838.363GB	SAS HDD	KPGYS6XF
☐	Controller Intel(R) VROC (Standard)				
☐	Non-RAID Drives				
☐	Drive 9	JBOD	1490.415GB	NVMe SSD	BTLN846006LX1P6AGN
☐	Drive 10	JBOD	1863.017GB	NVMe SSD	CVMD4134001N2P0JGN
☐	Drive 11	JBOD	1788.496GB	NVMe SSD	PHAB0133000L1P9SGN

Previous Next

OOB RAID configuration in UpdateXpress

Click each step in turn to see an example of OOB RAID configuration in UpdateXpress

Click each step in turn to see more information



Step



OOB RAID configuration in UpdateXpress

Select drives

Step



s UpdateXpress

1. Select Drives 2. Create Volume 3. Summary

Create the storage pool by specifying the RAID level and physical drives.

Select RAID Level:
RAID1 a minimum of 2 drive(s) for RAID 1. (*)

Available drives:

	Disk Drive	Type	Capacity
☐	Drive 2	HDD	136.732GB
☐	Drive 3	HDD	838.363GB

Add member ->
<- Remove
Add hot spare ->

Selected disk drives:

	Role	Disk Drive	Capacity
☐	Member	Drive 0	558.912GB
☐	Member	Drive 1	558.912GB

Previous Next Cancel

OOB RAID configuration in UpdateXpress

Create a volume

Step



UpdateXpress

1. Select Drives 2. Create Volume 3. Summary

Edit the volume property or more volumes if needed.

Create Volume Edit Volume Remove Volume

Volume Name	Capacity (GB)	RAID Level
-------------	---------------	------------

Free Storage Array Capacity:1117.824GB

Edit virtual Disk:

Virtual Disk Name: volume1

Capacity: 558.912 GB

Stripe Size: 256KB ▾

Read Policy: No Read Ahead ▾

Write Policy: Off ▾

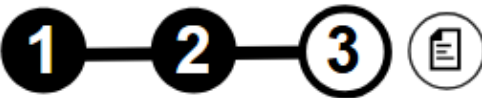
Apply Cancel

Previous Next Cancel

OOB RAID configuration in UpdateXpress

Summary

Step



UpdateXpress

1. Select Drives 2. Create Volume 3. Summary

Review the Summary and go back if you need to make corrections.

Storage Array:
RAID Level: 1
Number of Drives: 2
Number of Hot Spares: 0
Total Capacity: 1117.824 GB
Free Capacity: 558.912 GB

New Virtual Disks:

Volume Name	Capacity(GB)	RAID Level
volume1	558.912	RAID 1

Previous Create Cancel

One time boot configuration

The new `show boot` command is used to see the current system boot mode and boot order:

```
OneCli config show Boot --redfish
```

Possible output examples:

```
Boot.BootSourceOverrideEnabled=Disabled
```

```
Boot.BootSourceOverrideMode=UEFI
```

```
Boot.BootSourceOverrideTarget=None
```

```
Boot.UefiTargetBootSourceOverride=null
```

The `OneTimeBoot` command is used to change the boot mode or boot sequence only for the next boot.

In this example, the boot source is changed to the USB drive.

```
OneCli config batch --file OneTimeBoot.txt --redfish -b
```

```
USERID:PASSWORD@10.240.197.46
```

```
File OneTimeBoot.txt
```

```
set Boot.BootSourceOverrideEnabled Once
```

```
set Boot.BootSourceOverrideMode UEFI
```

```
set Boot.BootSourceOverrideTarget Usb
```

Collecting the Service Data Log with OneCLI

The new OneCLI `inventory` command can be used to collect the Service Data log (mini-log). The system must have XCC2 and OneCLI v4.0 or later.

```
OneCli.exe inventory getinfor --servicelog [--bmc userid:pass@bmchost]
```

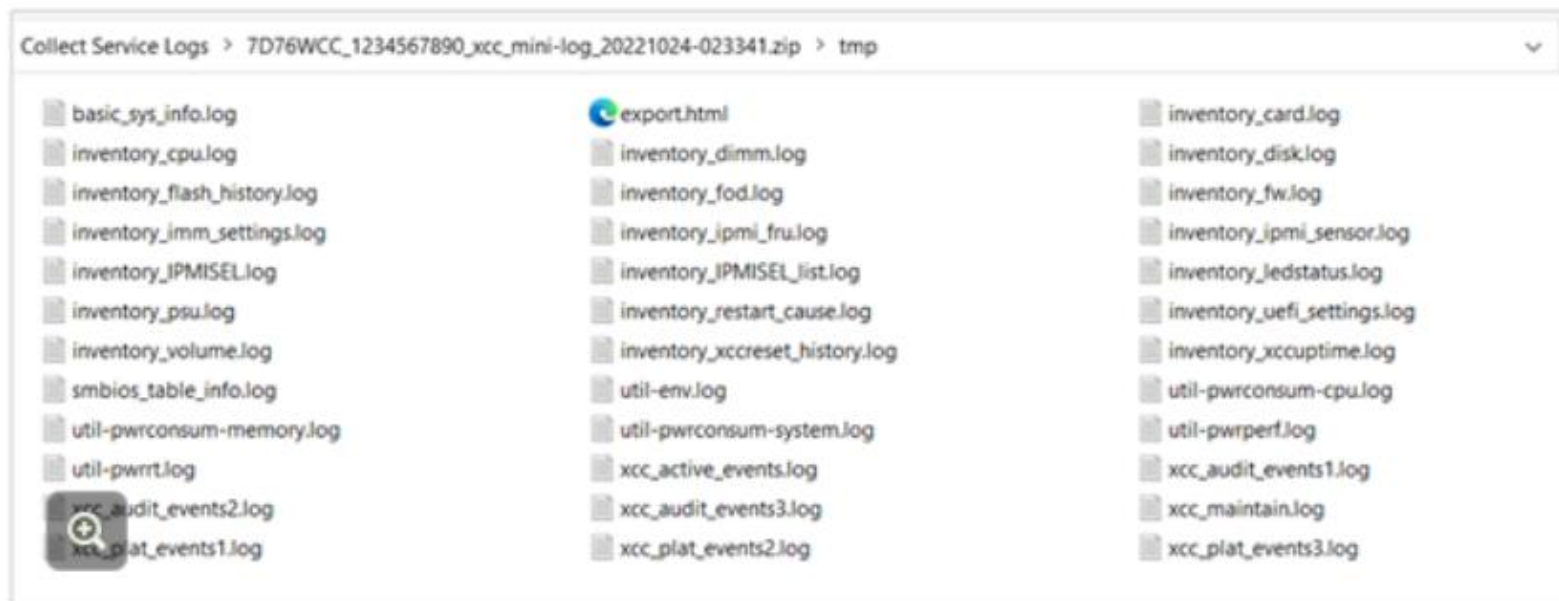
The OneCLI `misc` command can also be used to collect the mini-log

```
OneCli.exe misc servicelog [--bmc userid:pass@bmchost]
```

The naming convention for the Service Data Log file is:

MachineType+Model_SerialNumber_xcc_mini-log_date_time.zip – for example:

7D76WCC_1234567890_xcc_mini-log_20221024-023341.zip



Collecting service data with OneCLI

The OneCLI `misc servicedata` command, available from OneCLI 3.5 onwards, can be used to get service data information from the BMC.

- `Servicedata`
 - `--type/-T` to specify the type of service data information to get from the BMC – information types are as follows:
 - `osfailure`: to get the latest OS failure screen
 - `healthreport`: to get the health report
 - `all`: to get both the latest OS failure screen and the health report

Sample syntax:

```
OneCli.exe misc servicedata -type all -b USERID:PASSWORD@xxx.xxx.xxx.xxx
```

Example output destination:

```
\OneCli\bins\Invgy_util_lxce_onecli02c-4.0.0_winsrv_x86-64\servicedata\ServiceData-20220921-153132\
```

The latest OS failure screen is a picture saved as an XML file and zipped in a TGZ file – for example: *HealthReport-20220921-153132-56796.xml*

Miscellaneous new functions

Sysguard – for System Guard Service management

- `clearsnapshot` – to clear all snapshot data in BMC System Guard

Example syntax:

```
– ./OneCli sysguard clearsnapshot -b  
  USERID:PASSWORD@10.240.195.209
```

Hostinterface – Management Host Interface account

- `Clear` – to clear the Host Interface account

Example syntax:

```
– ./OneCli hostinterface clear --bmc-user USERID -bmc-password  
  PASSWORD
```


Miscellaneous new functions - continued

`rotkit` – to generate a boot kit for an RoT upgrade

- `--dir` – specifies the directory for system update packages (SUPs) - if no directory is specified, the current directory will be used
- `--rotmdir` – the RoT kit output directory – by default, `./rot_kit/` is used

```
D:\OneCli\bins\lnvgy_utl_lxce_onecli01w-4.0.0_winsrv_x86-64>OneCli.exe rotkit --dir D:\OneCli\EGS\UEFI --rotmdir rot_kit

Lenovo XClarity Essentials OneCLI lxce_onecli01w-4.0.0
(C) Lenovo 2013-2022 All Rights Reserved

OneCLI License Agreement and OneCLI Legal Information can be found at the following location:
"D:\OneCli\bins\lnvgy_utl_lxce_onecli01w-4.0.0_winsrv_x86-64\Lic"

Querying updates done.
Query results saved to: D:\OneCli\bins\lnvgy_utl_lxce_onecli01w-4.0.0_winsrv_x86-64\rotkit\Onecli-update-query.xml

Selected Packages:
=====
| No. | New Version | Machine Type | Update ID |
=====
| 1 | ESE107j-0.70 | 7D72, 7D73, 7D74, 7D75, 7D76, 7D77 | lnvgy_fw_uefi_esel07j-0.70_anyos_comp |
=====
Succeed.
```

Directory of `rot_kit` includes:

```
RoTFirmwareManifest.json
└─fw
    lnvgy_fw_uefi_esel07j-0.70_anyos_comp.uxz
```

More multi-function commands

Command	Description
multiflash	Update firmware requiring an upgrade on multiple remote BMC systems
multicompare	Compare firmware requiring an upgrade on multiple remote BMC systems
multiscan	Scan firmware requiring an upgrade on multiple remote BMC systems
multiconfig	Configure commands for multiple servers
multiinventory	Acquire system information from multiple servers
multisyshealth	Remotely collect system health from multiple systems
multiraid	Remotely configure RAID for multiple systems
multibmcpassword	Remotely change the BMC account password for multiple systems
multiospower	Operate the OS status – for example, turn on or off – for multiple servers
multiffdc	Get BMC/SMM FFDC from multiple servers
multiservicedata	Get service data information from multiple BMC accounts
multivm	Manage virtual medias on multiple BMC accounts